

The axiom of choice

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1 Introduction

Axiom 1.1 (Axiom of Choice) :

The Cartesian product of a non-empty family of non-empty sets is non-empty.

For a finite family of sets, we don't need this axiom, but if we want to generalize this principle to an infinite family of sets, we need it. Even if we think that it's intuitive, we can not do so without the Axiom of Choice, because if we want write all arguments, we would end up in a dead end.

So, to demonstrate you the importance of the Axiom of Choice, we will cite in a first part two equivalent formulations of it, after reminding some definitions. In a second part, we will show you that the Axiom of Choice implies Zorn's Lemma, which is our big part. We are concentrated about it. In a third part, we will demonstrate that Zorn's lemma implies the principle of Well-Ordering which implies the Axiom of Choice. And we will conclude with some examples of application of equivalents of the Axiom of Choice.

2 Vocabulary

Definition 2.1 :

Let X be a set. Let x, y be elements of X . Then " $\leq$ " is a **partial order** on X if and only if for all $x, y, z \in X$ we have

1. $x \leq x$, the Reflexivity.
2. If $x \leq y$ and $y \leq x$, then $x = y$, the Anti-Symmetry.
3. If $x \leq y$ and $y \leq z$, then $x \leq z$, the Transitivity.

Definition 2.2 (A chain) :

A subset B of a partially ordered set is a **chain** if and only if any two elements of B are comparable.
 B is a chain $\iff \forall x, y \in B, x \leq y$ or $y \leq x$.

Definition 2.3:

Let X be a partially ordered set and m be an element of X then we define

1. M is a **maximum** of $E \iff \forall x \in X, x \leq M$
2. M is a **maximal** element of $E \iff \forall x \in X, M \leq x \Rightarrow x = M$
3. m is a **minimum** of $E \iff \forall x \in X, m \leq x$
4. m is a **minimal** element of $E \iff \forall x \in X, x \leq m \Rightarrow x = m$

Definition 2.4 (A well-ordered set) :

A **well-ordered set** X is a set for which every non-empty subset has a minimum.

Definition 2.5 (Inductive) :

X is **inductive** if X is a partially ordered set such that every chain in X has an upper bound.

Definition 2.6 (Relation) :

Let E, F be sets, a relation $\mathcal{R}$ is an element of $\mathcal{P}(E \times F)$.

3 Axiom of Choice and equivalent formulations

The most famous formulation of the Axiom of Choice is the first (1.1), or this one that we called 1.bis

Axiom 3.1:

Let $(X_i)_{i \in I}$ a family of non-empty sets. Then, there exists a function of choice on the X_i , that's to say there exists an application c from I to the union $\bigcup_{i \in I} X_i$, such that for all $i \in I$, $c(X_i) \in X_i$.

So, in this part, we will show the equivalence between the Axiom of Choice, or 1.bis and (3.2) and (3.3).

Axiom 3.2:

Every relation includes a function with the same domain.

Axiom 3.3:

If $\mathcal{C}$ is a collection of pairwise disjoint non-empty sets, then there exists a set A such that $A \cap C$ is a singleton for each C in $\mathcal{C}$.

Proof :**AC $\Rightarrow$ Axiom 3.2:**

Let E, F be sets and $\mathcal{R} \in \mathcal{P}(E \times F)$ a relation. We shall remind that the domain of the relation $\mathcal{R}$ is defined as $Dom(\mathcal{R}) = \{x \in E \mid \exists y \in F, (x, y) \in \mathcal{R}\}$. By using the axiom of choice let f a choice function of $\mathcal{P}(F) \setminus \{\emptyset\}$. We can check that the following function works

$$\begin{aligned} g : Dom(\mathcal{R}) &\longrightarrow F \\ x &\longmapsto f(\{y \in F \mid (x, y) \in \mathcal{R}\}) \end{aligned}$$

Axiom 3.2 $\Rightarrow$ AC:

Let C be a set of non empty sets, we will define the set $E = \bigcup C$. We will also define the following relation

$$\mathcal{R} = \{(A, a) \in C \times E \mid a \in A\}$$

The axiom 3.2 then states that there exists a function f such that $Dom(f) = Dom(\mathcal{R})$ and $f \subset \mathcal{R}$. This function is a choice function for C .

AC $\Rightarrow$ Axiom 3.3:

Let C be a set of pairwise disjoint non-empty sets. Let f be a choice function for C . This means that

$$\forall X \in C, f(X) \in X$$

The following set has the wanted property

$$A = \{x \in E \mid \exists X \in C, x = f(X)\}$$

Axiom 3.3 $\Rightarrow$ AC:

Let C be a set of non empty sets. We will define $E = \bigcup C$ and

$$D = \{X \in \mathcal{P}(C \times E) \mid \exists A \in C, X = \{A\} \times A\}$$

Knowing that D is set of pairwise disjoint non-empty sets, the axiom 3.3 then gives us a set f such that $\forall X \in D, |f \cap X| = 1$. So then if $(A, x), (A, y) \in \{A\} \times A$ are such that $(A, x) \in f$ and $(A, y) \in f$ then we know that

$$\begin{aligned} (A, x) &\in (\{A\} \times A) \cap f \\ (A, y) &\in (\{A\} \times A) \cap f \end{aligned}$$

But we have that $|(\{A\} \times A) \cap f| = 1$, this means that

$$(A, x) \in f \text{ and } (A, y) \in f \Rightarrow x = y$$

We have proved that f is a function and is by definition a choice function.

□

4 Axiom of Choice $\Rightarrow$ Zorn's lemma

In this section we shall prove Zorn's lemma is a consequence of the Axiom of Choice. There are several different proofs of this lemma, the proof done here will be one of the most elementary proofs, by this we mean that this proof does not rely on any advanced tools.

The following definition will only be used in this section but is also useful in some branches of set theory.

Definition 4.1 (*Initial Segment*) :

Let $(E, \leq)$ be a partially ordered set we can then define what we call an initial segment of E by

$$\begin{aligned} A \in \mathcal{P}(E) \text{ is an initial segment of } E \\ \iff \\ \forall x \in A, \forall y \in E, y \leq x \Rightarrow y \in A \end{aligned}$$

The following definition defines a specific type of initial segment that will be used in our proof.

Definition 4.2:

Let $(E, \leq)$ be a partially ordered set and x be an element of E we define the interval $] -\infty; x[_E$ by

$$]-\infty; x[_E = \{y \in E \mid y < x\}$$

Proposition 4.3:

Let $(E, \leq)$ be well ordered set, let $I \in \mathcal{P}(E) \setminus \{E\}$ then

$$I \text{ is an initial segment of } E \Rightarrow \exists x \in E, I =] -\infty; x[_E$$

Proof :

Let us consider the set $A = E \setminus I$ we know that $I \neq E$, thus $A \neq \emptyset$. Since E is well ordered A has a minimum, we shall define it as x . Let $y \in E$ such that $y < x$ then $y \notin A$ so $y \in I$. This means that $] -\infty; x[_E \subset I$, the other inclusion being obvious we then have

$$]-\infty; x[_E = I$$

□

We will also define the following set which will only be used in this section.

Definition 4.4:

Let $(E, \leq)$ be a partially ordered set, we define:

$$\begin{aligned} \forall A \in \mathcal{P}(E), M(A) &= \{x \in E \mid \forall y \in A, y < x\} \\ F &= \{A \in \mathcal{P}(E) \mid A \text{ is a well ordered chain and } M(A) \neq \emptyset\} \end{aligned}$$

For each element A in F we know that $M(A) \neq \emptyset$, and we would like to choose for each A an element of $M(A)$ in order to define a function. But to do so we need the axiom of choice, in the following lemma note that the axiom of choice is used.

Lemma 4.5:

Let $(E, \leq)$ be a partially ordered set.

There exists a function $g \in E^F$ such that $\forall A \in F, g(A) \in M(A)$

Proof :

As mentioned earlier this lemma will use this axiom of choice to build g . Let us consider the following set $X = \mathcal{P}(E) \setminus \{\emptyset\}$. The elements of X are all non empty, this means that we can use the axiom of choice to obtain the following assertion

$$\exists f \in E^X, \forall A \in X, f(A) \in A$$

By noticing that $\forall A \in F, M(A) \in X$ we can define the following function

$$\begin{aligned} g : F &\longrightarrow E \\ A &\longmapsto f(M(A)) \end{aligned}$$

□

In the following the function g shall refer to the one that we defined in the lemma above. The following definition is specific to the proof of Zorn's lemma.

Definition 4.6:

Let $(E, \leq)$ be a partially ordered set, let $A \in \mathcal{P}(E)$ we shall say that

$$A \text{ is a } g\text{-chain} \iff A \in F \text{ and } \forall x \in A, x = g(\text{--}\infty; x|_A)$$

Proposition 4.7:

Let $(E, \leq)$ be a partially ordered set such that for every well ordered chain A , $M(A) \neq \emptyset$. Let $A \in \mathcal{P}(E)$, if A is a g-chain then $A \cup \{g(A)\}$ is also a g-chain.

Proof :

Let's start by showing that $A \cup \{g(A)\}$ is a well ordered chain. We shall define $B = A \cup \{g(A)\}$. Let $N \in \mathcal{P}(B)$.

- If $N \cap A = \emptyset$ then $N = \{g(A)\}$ and we then notice that $g(A)$ is a minimum of this set.
- If $N \cap A \neq \emptyset$ then by knowing that A being a g-chain is well ordered and that $N \cap A \in \mathcal{P}(A) \setminus \{\emptyset\}$ we can deduce that $N \cap A$ has a minimum m . And we also have, by definition of $g(A)$ that $m \leq g(A)$. So m is a minimum of $N = (N \cap A) \cup (N \cap \{g(A)\})$.

This shows that B is well ordered, and B is well ordered $\Rightarrow M(B) \neq \emptyset \Rightarrow B \in F$, thus $g(B)$ is well defined. Now let's show that B is a g-chain.

Let $x \in B$.

- If $x = g(A)$ we just need to notice that $A =]-\infty; g(A)[_B$ because $g(A)$ is a strict maximal element and A is well ordered. Thus we have $g(A) = g(]-\infty; g(A)[_B)$
- If $x \neq g(A)$ then $x \in A$ and $x < g(A)$, since A is a g-chain we have $x = g(]-\infty; x[_A)$. And $x < g(A) \Rightarrow]-\infty; x[_A =]-\infty; x[_B$, thus $x = g(]-\infty; x[_A) = g(]-\infty; x[_B)$.

We have showed that

$$\forall x \in B, x = g(]-\infty; x[_B)$$

Adding this to the fact that B is well ordered we know that B is a g-chain.

□

Lemma 4.8:

Let $(E, \leq)$ be a partially ordered set, let A, B be g-chains of E , then A is an initial segment of B or B is an initial segment of A .

Proof :

Let us define the following set $X = \{J \in \mathcal{P}(E) \mid J \text{ is an initial segment of } A \text{ and } B\}$, this set is non empty because $\emptyset \in X$. We can now consider $I = \bigcup X$. Let us notice that I is again an initial segment of A and B . Let us suppose that $I \neq A$ and $I \neq B$, by using the proposition 4.3 we then have

$$\begin{aligned} \exists a \in A, I &=]-\infty; a[_A \\ \exists b \in B, I &=]-\infty; b[_B \end{aligned}$$

Knowing that A and B are g-chains we then have $a = g(\cdot - \infty; a|_A) = g(I) = g(\cdot - \infty; b|_B) = b$, and this means that $I \cup \{g(I)\}$ is an initial segment of A and B . But this means that $I \cup \{g(I)\} \in X$ and $I \cup \{g(I)\} \subset I$, which is impossible so, we either have $I = A$ or $I = B$.

□

Lemma 4.9:

Let $(E, \leq)$ be a partially ordered set.

There exists $A \in \mathcal{P}(E)$ a well ordered chain such that $M(A) = \emptyset$.

Proof :

Let us consider the set $G = \{A \in \mathcal{P}(E) \mid A \text{ is a g-chain}\}$, this set is non empty because $\emptyset \in G$. We can then consider $C = \bigcup G$. Let's show that $M(C) = \emptyset$, to do so, we will do a proof by contraposition. First lets prove that every g-chain is an initial segment of C .

Let A be a g-chain, let $a \in A$ and $b \in C, b \leq a$. We know that $b \in C \Rightarrow \exists B \in G, b \in B$. A and B are g-chains so we know that A is an initial segment of B or B is an initial segment of A . In both cases $b \in A$. This proves that A is an initial segment of C . We can now show that C is a well ordered chain.

Let $N \in \mathcal{P}(C) \setminus \{\emptyset\}$.

Let $a \in N$, we then know that $\exists A \in G, a \in A$. Since A is a g-chain, A is well ordered and we can notice that $A \cap N \in \mathcal{P}(A) \setminus \{\emptyset\}$, which means that $A \cap N$ has a minimum m . A is also an initial segment of C so let $b \in N$. Then $b \geq a \Rightarrow b \geq m$, and $b \leq a \Rightarrow b \in A \Rightarrow b \in A \cap N \Rightarrow m \leq b$. So N has a least element which is m .

Lets also show that $M(C) \neq \emptyset \Rightarrow C$ is a g-chain, let $a \in A$, then $\exists A \in G, a \in A$. Since A is a g-chain we have $a = g(\cdot - \infty; a|_A)$. We can also notice that A is an initial segment of C this implies that $\cdot - \infty; a|_A = \cdot - \infty; a|_C$. Thus, we have $a = g(\cdot - \infty; a|_A) = g(\cdot - \infty; a|_C)$, so $M(C) \neq \emptyset \Rightarrow C$ is a g-chain, and we know that $C \cup \{g(C)\} \notin G$ so is not a g-chain. But if a set A is a g-chain, then by using the proposition 4.7 $A \cup \{g(A)\}$ is a g-chain. So this means that C is not a g-chain, which means that $M(C) = \emptyset$ by contraposition.

□

Lemma 4.10 (Zorn version 1) :

Let $(E, \leq)$ be a partially ordered set.

If every well ordered chain in E has an upper bound then $\exists M \in E, M$ is a maximal element.

Proof :

Lets do a proof by contraposition. We shall show that

$$\forall M \in E, M \text{ is not a maximal element} \Rightarrow \forall A \in \mathcal{P}(A), (A \text{ is a well ordered chain} \Rightarrow M(A) \neq \emptyset)$$

Let A be a well ordered chain then we know that $\exists M \in E, \forall a \in A, a \leq M$, but M is not a maximal element so $\neg(\forall x \in X, M \leq x \Rightarrow x = M)$ and we also have that

$$\begin{aligned} \neg(\forall x \in X, M \leq x \Rightarrow x = M) &\Rightarrow \exists x \in X, \neg(M \leq x \Rightarrow x = M) \\ &\Rightarrow \exists x \in X, M \leq x \text{ and } x \neq M \end{aligned}$$

Let x be such an element, since A is a well ordered chain, we then have $\forall a \in A, a < x$. This means that $x \in M(A)$ and so $M(A) \neq \emptyset$. By using the lemma 4.9 and the contraposition of the statement that we proved we know that E must have a maximal element.

□

Corollary 4.11 (*Zorn version 2*) :

Every inductive set has a maximal element

Proof :

By noticing that a well ordered chain is a chain we can use the lemma 4.10 to conclude.

□

5 Zorn's lemma $\Rightarrow$ Well-Ordering $\Rightarrow$ Axiom of Choice

Proposition 5.1 :

The principle of well-ordering: Every set can be well ordered.
Zorn's lemma $\Rightarrow$ Well-ordering

Remark. The principle of Well-ordering is also called Zermelo's Theorem, because this mathematician worked about it.

Proof :

Let X a set. Let $\Omega = \{Y \subset X \mid Y \text{ can be well-ordered}\}$, we can do this because a singleton is by definition well-ordered. Let $\preceq$ the relation "to be prolonged by" on the orders of all $Y \in \Omega$. It is an order, inductive, because each chain is dominated by her union. So, by Zorn's lemma, there exists a maximal element, call it $(\bar{Y}, \preceq)$ for $\preceq$.

If $\bar{Y} \subset X$, ($\bar{Y} \neq X$) there exists $a \in X \setminus \bar{Y}$. So we can define a good order on $\bar{Y} \cup \{a\}$, preserving the same order for the elements of $\bar{Y}$, and saying $y \leq a$ for all $y \in \bar{Y}$. But this good order prolongs that one on $\bar{Y}$. so we have a contradiction with maximality.

So $\overline{Y} = X$ and the good order on $\overline{Y}$ is a good order on X .

□

Proposition 5.2:

Well-ordering $\Rightarrow$ Axiom of Choice

Proof :

Let C be a set of non empty sets, by using the well ordering theorem we will create a choice function on this set, thus proving that the theorem of well ordering implies the axiom of choice.

Let us define $E = \bigcup C$. The theorem of well ordering implies that there exists an order $\leq$ on E such that $(E, \leq)$ is well ordered. For $A \in \mathcal{P}(E)$ we will define $\min(A) = \{a \in A \mid \forall x \in A, a \leq x\}$. The set being well ordered means that we can define the following set

$$f = \{(A, a) \in C \times E \mid a \in \min(A)\}$$

Let $A \in C$, E is well ordered $\Rightarrow \exists a \in \min(A)$, thus $(A, a) \in f$. We have shown that

$$\forall A \in C, \exists x \in E, (A, x) \in f$$

Let $(A, x) \in f$ and $(A, y) \in f$.

We then have

$$x \in \min(A)$$

$$y \in \min(A)$$

Thus we have $x \leq y$ and $y \leq x$ by definition of $\min(A)$, this means that $x = y$. So we have showed that

$$\forall A \in C, \forall x, y \in E, ((A, x) \in f \text{ and } (A, y) \in f) \Rightarrow x = y$$

This means that f is a function. And f is a choice function because

$$\forall A \in C, f(A) \in \min(A) \subset A$$

□

6 Examples

Theorem of existence of a basis and uncompleted basis

Theorem 6.1 (*Theorem of existence of a basis and of uncompleted basis*) :

Let E be a vector space and let L be a linearly independent set, then there exists a basis B containing L .

Proof :

Let's consider the set $X = \{F \in \mathcal{P}(E) \mid F \text{ is a linearly independent set containing } L\}$. This set is inductive using the order $\subset$, so by using Zorn's lemma there is a maximal element that we will denote as B . Let's show that B is a basis. If B is not a basis then there exists an element x such that $x \notin \text{Vect}(B)$ thus $B \cup \{x\}$ is a linearly independent set. But this contradicts the maximality of B , so B must be a basis.

□

Theorem of Hahn-Banach

We talk about this theorem because we studied this in the first semester in Analysis, and in his proof, we use Zorn's Lemma without to demonstrate it. This theorem has a lot of versions and his proof is long, so we give just the big ideas and when we use Zorn's Lemma.

Theorem 6.2 (*Theorem of Hahn-Banach*) :

Let E a vector space, p a semi-norm on E . Let M , a sub vector space of E . Let $f : M \rightarrow \mathbb{R}$ with f linear such that $\forall x \in M \implies f(x) \leq p(x)$.
Then there exists $F : E \rightarrow \mathbb{R}$ linear which prolongs f and such that $\forall x \in E, F(x) \leq p(x)$.

Proof :

Let $A = \{(h, M_h) : M_h \text{ is a sub vector space of } E, M \subset M_h, h : M_h \rightarrow \mathbb{R} \text{ linear and } h|_M = f \text{ and } h(x) \leq p(x) \forall x \in M_h\}$.

We remark that A is non-empty. We define $\preceq$, a partial order by

$$(h, M_h) \preceq (g, M_g) \text{ if } M_h \subseteq M_g \text{ and } g|_{M_h} = h$$

We prove A is inductive and M_g is a sub vector space. By Zorn's Lemma A has a maximal noted (F, M_F) . By contraposition we show that $M_F = E$.

□

Besides, some of other projects need to use the Axiom of Choice or Zorn's Lemma or the principle of Well-Ordering to establish certain results.